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Foundations of Modern Hypothesis Testing: A Comprehensive Analysis of Neyman & Pearson (1933)

Statistical Seminar Group One

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- Statistical inference seeks conclusions from sample data under uncertainty.
- Early 20th century: hypothesis testing lacked a unified mathematical framework.
- Neyman and Pearson (1933) introduced:
 - Null and alternative hypotheses.
 - Type I and Type II errors.
 - Significance level and power.
 - Likelihood ratio as optimal decision criterion.
- Their work transformed hypothesis testing into an optimization problem.

Historical Background and Motivations

- Prior to 1933, researchers relied on probability calculations without formal decision rules.
- Neyman and Pearson proposed evaluating procedures by long-run error rates.
- Key innovations:
 - ① Formal distinction between Type I and Type II errors.
 - ② Power as probability of detecting false nulls.
 - ③ Most powerful tests via likelihood ratio.
- Unified many existing tests (t, F, chi-square) under a common theory.

The Neyman-Pearson Lemma

Problem

Test simple null $H_0 : f(x) = f_0(x)$ vs. $H_1 : f(x) = f_1(x)$ with significance level α .

Lemma

The most powerful test rejects H_0 if the likelihood ratio $\frac{f_1(x)}{f_0(x)} > k$, where k is chosen so that $\Pr(\text{reject } H_0 \mid H_0) = \alpha$.

- Maximizes power for a given α .
- Provides theoretical foundation for optimal tests.

Extension to Composite Hypotheses

- Composite hypotheses involve unknown nuisance parameters.
- Need for *similar critical regions* to maintain constant α .
- Example: Student's t -test for testing $\mu = \mu_0$ when variance unknown.
- $t = \frac{\sqrt{n}(\bar{x} - \mu_0)}{s}$ follows t_{n-1} under H_0 .
- Critical value chosen such that $\Pr(T > t_0) = \alpha$.

Assumptions Underpinning the Framework

- ➊ **Simple Hypotheses:** Both distributions fully specified.
- ➋ **i.i.d. Observations:** Independent and identically distributed.
- ➌ **Correct Model Specification:** Assumed distributions match reality.
- ➍ **Fixed Significance Level:** α chosen a priori.
- ➎ **Likelihood Ratio as Optimal Criterion:** Based on the Neyman-Pearson lemma.

In practice, these may be relaxed or extended (e.g., using Welch's t-test when variances unequal).

Data and Hypothesis

Iris Dataset (Fisher, 1936)

- 150 observations of three Iris species.
- Subset: Setosa and Versicolor (100 obs, 50 each).
- Variables: Sepal Length, Sepal Width, Petal Length, Petal Width.

Hypothesis

$H_0 : \mu_{\text{Setosa}} = \mu_{\text{Versicolor}}$ vs. $H_1 : \mu_{\text{Setosa}} \neq \mu_{\text{Versicolor}}$. Significance level $\alpha = 0.05$.

Table: Summary Statistics for Sepal Length (cm)

Statistic	Setosa	Versicolor
Mean	5.006	5.936
Median	5.000	5.900
Min	4.300	4.900
Max	5.800	7.000

- Boxplots show clear separation: Versicolor has longer sepals.
- Preliminary evidence of a difference.

Assumption Checks

Table: Summary of Assumption Tests

Assumption	Test	Statistic	p-value
Normality (Setosa)	Shapiro-Wilk	W=0.9777	0.4595
Normality (Versicolor)	Shapiro-Wilk	W=0.9778	0.4647
Equal Variance	F-test	F=0.4663	0.0087
Independence	Durbin-Watson	DW=1.7617	0.0959

- Normality satisfied.
- Variances unequal ($p = 0.0087$) $\Rightarrow$ use Welch's t-test.
- Independence satisfied ($DW \approx 2$, $p=0.0959$).

Welch's Two-Sample t-Test Results

Table: Hypothesis Test Results

Test	t-statistic	df	p-value
Two-sided	-10.521	86.54	$< 2.2 \times 10^{-16}$
Left-tailed	-10.521	86.54	$< 2.2 \times 10^{-16}$
Right-tailed	-10.521	86.54	1.000

Decision

Reject H_0 in two-sided and left-tailed tests.

Conclusion

Mean sepal length differs significantly; Setosa has shorter sepals on average.

Critical Regions

- With $\alpha = 0.05$ and $df \approx 86.54$:
 - Two-sided critical values: ± 1.9878
 - Left-tailed critical value: -1.6627
 - Right-tailed critical value: 1.6627
- Observed $t = -10.521$ falls in all critical regions except the right-tailed.

Implication

The observed statistic is far into the rejection region, providing very strong evidence against H_0 .

Power Analysis

- Effect size (Cohen's d) = 2.104 (very large).
- Power with $n = 50$ per group: $\sim 100\%$.
- Required n for 80% power: only about 5 per group.

Interpretation

- Current sample size is more than sufficient.
- Risk of Type II error is negligible.
- Large effect ensures high detectability.

Likelihood Ratio Test (Illustration)

- For Setosa sepal length, tested $H_0 : \mu = 5.0$ vs. $H_1 : \mu = \bar{x} = 5.006$.
- Likelihood under H_0 : 1.1087×10^{-8} .
- Likelihood under H_1 : 1.1168×10^{-8} .
- Likelihood ratio $L_0/L_1 = 0.9928 < 1$.

Interpretation

Data are more likely under H_1 , consistent with the t-test results. The evidence favours the alternative, even though the difference is small.

Type I and Type II Errors

Table: Decision Outcomes

Reality	Decision	Outcome
H_0 True	Reject H_0	Type I Error ($\alpha = 0.05$)
H_0 False	Fail to reject	Type II Error (β)
H_0 False	Reject H_0	Correct (Power = $1 - \beta$)

- In our analysis, p-value is extremely small, so Type I error risk is negligible.
- Power is almost 1, so Type II error risk is virtually zero.

Strengths and Weaknesses

Strengths

- Mathematical rigour.
- Decision-theoretic foundation.
- Unification of many tests.
- Guaranteed long-run error rates.

Weaknesses

- Requires well-defined alternatives.
- Composite hypotheses complexity.
- One-sided bias.
- Arbitrary choice of α (e.g., 0.05).
- Does not incorporate prior information.

Comparison with Bayesian and ML Methods

Neyman-Pearson	Bayesian	Machine Learning
Controls error rates	Posterior probabilities	Focus on prediction
Fixed α	Prior required	No formal inference
Decision rule	Probability of hypothesis	Black-box
Assumptions matter	Prior sensitivity	More robust to violations

- Choose method based on research question, data, and interpretability needs.

Recommendations for Modern Application

- 1 Use computational tools (R, Python) for exact p-values and power calculations.
- 2 Adjust for multiple testing when necessary.
- 3 Report effect sizes and confidence intervals alongside p-values.
- 4 Validate assumptions with modern diagnostics (bootstrapping, permutation tests).
- 5 Ensure reproducibility: share code, data, and documentation.
- 6 Consider equivalence tests when relevant.

- The Neyman-Pearson framework provided a rigorous foundation for hypothesis testing.
- Key contributions: error control, power, and optimal tests via likelihood ratio.
- Implementation on Iris dataset demonstrated:
 - Clear difference between Setosa and Versicolor sepal lengths.
 - Proper assumption checking and test selection.
 - High power and negligible error rates.
- Despite limitations, the principles remain central to statistical practice.
- The paper's ideas continue to influence modern inference and machine learning.

References

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Code and data available at: <https://github.com/pewandera/Neyman-Pearson-Hypothesis-Testing>